

UDBHAV SRIVASTAVA

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Education

Texas A&M University, MS in Data Science, GPA:3.67/4.0 May 2024

Courses: Machine Learning, Deep Learning, Data Mining, Predictive Analytics, Statistics, Artificial Intelligence

MIT World Peace University India, Btech in Computer Science, GPA:9.22/10.0 May 2022

Courses: Big Data, Data Science, Cloud Computing, Data Warehouse, Data Management, Business analytics

Experience

GeoSAT, Texas A&M University July 2024 – Present

Data Science Researcher

College station, TX

- Benchmarked **LLAMA-3** model on location extraction from tweets (F-1 score of 0.92), improving the accuracy of location identification by 10%, enabling faster crisis response efforts.
- Leveraged Low-Rank Adaptation (**LORA**) & Parameter-Efficient Fine-Tuning (**PEFT**) techniques, reducing computational costs and allowing for faster training and improved performance.
- Outperformed traditional transformer models (**BERT, RoBERTa**) by 2% in F1-score, demonstrating the superior accuracy and effectiveness of the LLAMA-3 model in extracting location information from crisis-related tweets.

IEEE Move Global, Texas A&M University Aug 2023 – Jan 2024

Capstone Project

College station, TX

- Developed rule-based & ML model (**K-means Clustering**), predicting optimal locations for deploying truck in disaster scenarios, leading to cost savings of \$35,000 per deployment.
- Deployed predictive models (**Linear & Random Forest Regression**) to forecast resource needs (power, network, fuel etc) based on historical real-time sensor readings, reducing resource waste by 10x.
- Maintained databases(**MySQL**) to manage the collected data for efficient access and analysis, reducing data retrieval time by 60%, which enabled improvement in emergency response times.
- Developed data visualizations(**Tableau**) to identify network coverage patterns and optimize deployment strategies, resulting in a 40-mile expansion of overall communication coverage in critical disaster zones.

IDEaS - a SAS Company Jul 2021 – Feb 2022

Data Analyst Intern

Pune, India

- Constructed advanced sales forecasting models (**ARIMA, LSTM, VAR**) to predict the next quarter's sales, leading to improved budget planning and increased profit by \$60,000.
- Developed an end-to-end data pipeline for the COVID-19 relief system, utilizing **AWS (S3, Lambda)**, enabling faster data processing and timely support to 10,000+ chains of clients.
- Implemented an automated pipeline utilizing **REST APIs & SQL** to scrape data & store in database, increasing data collection efficiency and reducing manual work of 90 hrs per week
- Designed insightful visual report **PowerBI** and employed **DAX** calculations to analyze product interventions, enabling informed strategic decisions, & contributing to cost savings of \$12,000 weekly.
- Leveraged **MS Excel** to develop a comprehensive accounts budget planning system, providing managers with real-time scenario planning, resulting in a 27% improvement in ROI.

Projects

AnalyticaGPT - LLM for Data Analysis | [WebApp](#) | [Github](#)

- Engineered an AI web application using **GPT-4 LLM** to input user's dataset and perform Exploratory Data Analysis (EDA) tasks, eliminating the need for users to write code.
- Employed **LangChain** agent to automatically generate and execute Python commands based on user's prompt, streamlining the process of data analysis (**Pandas**) and visualization (**Matplotlib, Plotly, Seaborn**)

SoccerVision - Yolo Object Detection | [Github](#)

- Leveraged **Computer Vision** to decompose video files into frames and Implemented **YOLOv5** model using **pytorch & Cv2** for real-time object detection (players, referees, ball) in user-uploaded videos, applying transfer learning.
- Utilized **K-means clustering** to automatically identify teams based on jersey colors, enabling accurate ball possession analysis & data-driven game analysis.

Chatbot & NLP for Suicide Risk Detection | [Website](#) | [Github](#)

- Built a high-accuracy (97%) NLP classification model (**Pytorch**) for suicide risk detection, leveraging models [**CNN, LSTM, BERT, ELECTRA**], enabling the potential for improved mental health support systems.
- Cleaned & Preprocessed 250k+ record dataset for suicide risk detection using **Spacy, TextBlob, and NLTK**.

AWS-Powered Ethereum Forecasting | [WebApp](#) | [Github](#)

- Forecasted Ethereum prices using **TimeGPT**, a SOTA time series model with RMSE reduced by 61% compared to base model.
- Hosted a scalable web application on **Amazon Elastic Compute Cloud (EC2)** AWS Route 53 for cost effective infrastructure.

Sales-Trend Data Analysis | [Website](#) | [Dashboard](#) | [Github](#)

- Leveraged **SQL window functions** to analyze cumulative sales trends, providing critical insights on seasonality, empowering datadriven decision-making for inventory optimization by 17%.
- Developed a robust **Tableau** dashboard to evaluate historical profit and loss patterns, enhancing decision-making for pricing strategies and product development.

Skills

Languages: Python, C/C++, HTML/CSS, JavaScript, SQL, Bash, Unix, SAS, VBA, Vlookup, Scala.

Tools: Azure, MS Office, Docker, Github/Git, Databricks, Pyspark, Hugging Face, Transformers, Attention models.

Libraries: Pytorch, Tensorflow, Keras, Pandas, Numpy, Scikit-Learn, Scipy, Matplotlib, Seaborn, Plotly, Prophet, NTLK, Spacy.

Competencies: Large Language Model, Natural Language Processing, Computer Vision, Hypothesis Testing, A/B testing.

Soft Skills: Effective Communicator, Agility, Decision Making, Problem-Solving, Analytical Thinker, Adaptable, Detail-Oriented.